

GlaciArc

Glacier Risk Prediction and Monitoring Platform

Project Team

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Chapter 1

Introduction

GlaciArc, is web application designed for monitoring glaciers in Gilgit Baltistan and predicting risks related to glacier lake outburst floods (GLOFs) through satellite imagery analysis. This document is intended for various readers, including developers, project managers, marketing staff, end-users, and testers.

1.1 Problem Statement

The frequency of Glacier Lake Outburst Floods (GLOFs) has been reported to have increased by over 23% over the course of the last 30 years, endangering over 7.1 million people in the northern regions of Pakistan. [1].[2]. Despite this eminent problem the current monitoring systems prove to be inadequate when it comes to predicting short term risks such as floods and tracking glacier movements. GlaciArc aims to overcome this by integrating satellite image monitoring and advanced machine learning algorithms to provide timely risk assessment and early warning to improve disaster preparedness.

1.2 Scope

GlaciArc aims to develop a user friendly, web-based monitoring and predictive platform for glacier related hazards in Pakistan's northern regions particularly focused on Gilgit Baltistan. By leveraging satellite imagery and advanced machine learning models, the platform will track glacier dynamics, predict potential GLOF events and evaluate impacts on nearby rivers, roads, and buildings. The primary functionalities of GlaciArc would involve analyzing how changes in glaciers alter river courses, detect signs of land loss, and classify surroundings into risk zones based on their susceptibility to glacier-related hazards. GlaciArc will provide clear insights, so relevant authorities and concerned com-

munities can better understand the risk and prepare accordingly. GlaciArc will also utilize a diverse dataset with a wide array of features such as roads, buildings, glaciers, rivers, and green lands to ensure definite evaluations and risk categorization. The platform aspires to become an essential tool for enhancing disaster preparedness and safeguarding communities from glacier-induced disasters.

1.3 Modules

1.3.1 Data Collection

This module focuses on collecting and organizing the data required for our project. It involves gathering satellite images of the concerned regions and then processing said data to create a usable dataset.

1. **Satellite Image Acquisition:** Collect satellite images of glaciers and surrounding areas of Gilgit Baltistan for monitoring and analysis.
2. **Data Processing:**
 - Processing the collected data to create a usable dataset.

1.3.2 Model Evaluation

This module will focus on selecting and training machine learning models to analyze glacier dynamics and their impacts using the dataset prepared in Module 1.

1. **Model Selection:**
 - Choose appropriate machine learning models such as U-Net, Random Forest (RF), and YOLO that align with the requirements of the project.
 - Evaluate the selected models for their ability to deal with large datasets.
2. **Model Training:**
 - **Prepare Training Data:** Focus on organizing the data in a suitable form for training, such as splitting it into training and validation sets.
 - **Model Training:** Train the selected model on the dataset by feeding it data and allowing it to learn different patterns.
 - **Monitor Progress:** Track parameters such as loss and accuracy while training to ensure that the models are learning effectively.

1.3.3 Testing

This module emphasizes evaluating the performance of the trained machine learning models to ensure they meet the required standards of the project.

1. Model Evaluation:

- Assess the models using performance metrics to determine their effectiveness and efficiency.

2. Performance Assessment:

- Identify any errors in the model predictions and adjust accordingly.

3. Comparative Analysis of Models:

- Analyze the models based on certain evaluation metrics (accuracy, precision, etc.).

1.3.4 Web Application Development

This module focuses on creating a user-friendly and visually appealing web application, ensuring a functional and effective platform for the project.

1. UI Design:

- **Interface Designing:** Focus on designing an intuitive, user-friendly interface for the web application.
- **User Experience:** Ensure the design is easy to use and navigate for the intended audience.
- **Data Visualization Integration:** Design the interface to include a live map feature where users can view and interact with geospatial data, including glaciers, rivers, and infrastructure.

2. Web App Development:

- **Building Web Platform:** Develop the web-based platform and integrate machine learning models and data analysis features.
- **Integration:** Ensure seamless integration of the user interface with the machine learning models for smooth implementation.

- **Map Visualization:** Integrate libraries like leaflet.js, ipyleaflet, or folium to visualize geospatial data on interactive maps. This includes coding for map interactions and ensuring smooth integration with the existing web application.
- **Data Display:** Ensure the map displays up-to-date information from the processed datasets and provides users with options to explore different features, such as zooming in on specific areas or viewing detailed risk assessments.

3. Testing and Deployment:

- **Functional Testing:** Focus on ensuring all features work as intended in the web app and are easy to use.
- **Deployment:** Deploy the web application to a production environment, making it accessible to users.

1.4 User Classes and Characteristics

Table 1.1: User Classes and Descriptions for GlaciArc System

User Class	Description
Researcher	A Researcher is a user focused on analyzing glacier data and predicting glacier movements. They will access satellite images, process data, and view predictions to support their research.
Administrator	The Administrator is responsible for managing user accounts, overseeing system performance, and ensuring data integrity.
End User	An End User is any individual who wants to view glacier data and predictions. They will select specific glaciers, adjust settings, and visualize results on the dashboard.
Stakeholder	Stakeholders are individuals or organizations interested in the outcomes of the GlaciArc project, including government officials, NGOs, and community leaders.

1.5 Research on Existing Projects

Project Name	Description	Features	Research Gap/Uniqueness of GlaciArc
NASA's Earth Science GLOF Prediction	Uses satellite data and machine learning models to predict GLOFs in specific regions.	Focuses on satellite data for GLOF prediction.	GlaciArc combines GLOF predictions with risk analysis for nearby infrastructure and interactive live mapping.
Swiss Glacier Monitoring Network	Monitors glacier dynamics and changes in ice mass over time using satellite imagery.	Provides glacier mass balance and volume change assessments.	GlaciArc integrates monitoring with predictive models for GLOF and infrastructure hazard zones.
HIMAP Glacier Dynamics Project	Studies glacier retreat and its impact on water resources in the Himalayas.	Focuses on glacier retreat and water availability.	GlaciArc extends to predicting glacier-related risks (e.g., GLOF) and evaluating effects on infrastructure and nearby communities.

Table 1.2: Research on Existing Projects

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case

2.1.1 Use Case 1: Select Glacier for Analysis

Actor: User (Researcher/General User)

Description: The user selects a specific glacier to analyze and view relevant data.

Preconditions: The user must be logged into the GlaciArc system.

Postconditions: The selected glacier is displayed with available options for analysis.

Main Flow:

1. The user navigates to the "Select Glacier" section of the application.
2. The user browses list of glaciers.
3. The user selects the desired glacier (e.g., Shisper Glacier).
4. The system displays a summary of available data for the selected glacier.

Alternative Flows:

- If the selected glacier has no available data, the system informs the user and suggests alternatives.

2.1.2 Use Case 2: Adjust Year for Data Viewing

Actor: User (Researcher/General User)

Description: The user changes the year to view data related to glacier conditions and risks.

Preconditions: A glacier must be selected.

Postconditions: The user views data corresponding to the selected year.

Main Flow:

1. The user selects the "Year" dropdown menu.
2. The user chooses a specific year from the list (e.g., 2022).
3. The system retrieves data for the selected year and updates the display.
4. The system presents the relevant information, including images and risk assessments.

Alternative Flows:

- If data for the selected year is unavailable, the system alerts the user.

2.1.3 Use Case 3: View Predictions and Risk Zones

Actor: User (Researcher/General User)

Description: The user views predictions and risk zones for the selected glacier and year.

Preconditions: A glacier must be selected, and a year must be set.

Postconditions: The user can see visualized data including predictions and danger zones.

Main Flow:

1. The user navigates to the "View Predictions" section.
2. The user clicks on the "Display Predictions" button.
3. The system retrieves the predictions for the selected glacier and year.
4. The system visualizes the results on a map, marking safe and danger zones.

5. The user can interact with the visualization to explore detailed data.

Alternative Flows:

- If predictions fail to load, the system notifies the user and prompts them to retry.

2.1.4 Use Case 4: Adjust Visualization Settings

Actor: User (Researcher/General User)

Description: The user modifies the visualization settings to enhance the display of predictions.

Preconditions: A glacier must be selected, and predictions must be displayed.

Postconditions: The visualization settings are updated based on user preferences.

Main Flow:

1. The user accesses the "Settings" or "Visualization Options" menu.
2. The user adjusts parameters (e.g., labels, map layers).
3. The user clicks on the "Apply" button.
4. The system updates the visualization according to the new settings.

Alternative Flows:

- If the settings lead to an error in display, the system reverts to default settings and informs the user.

2.1.5 Use Case Diagram

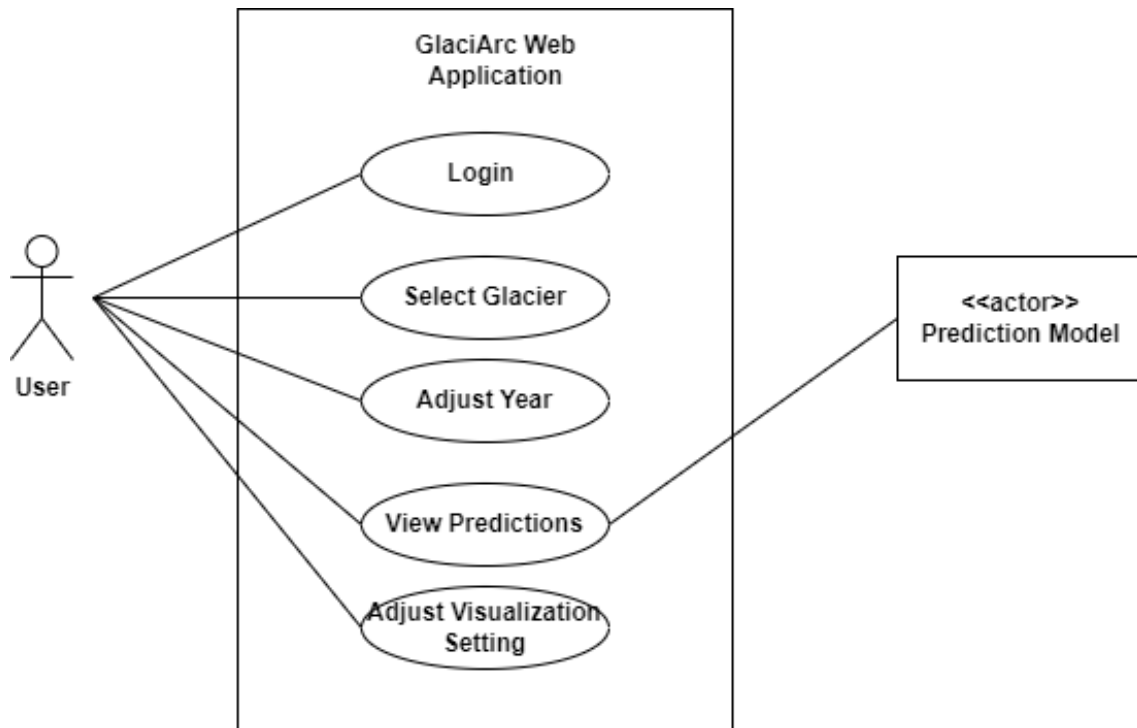


Figure 2.1: Use case diagram

2.2 Functional Requirements

2.2.1 Module 1

Following are the requirements for module 1:

1. FR1: The system should be able to acquire satellite imagery data for specified glacier areas in Gilgit Baltistan.
2. FR2: The system should be able to preprocess the collected satellite imagery data to create a usable dataset for analysis.
3. FR3: The system should be able to crop the satellite images to focus on specific glacier regions of interest.
4. FR4: The system should be able to label the processed data.
5. FR5: The system should be able to store the processed dataset in an organized manner for future access and analysis.

2.2.2 Module 2

Following are the requirements for module 2:

1. FR1: The system should be able to select appropriate machine learning models, such as U-Net, Random Forest, and YOLO, that align with the project requirements.
2. FR2: The system should be able to prepare the training data by organizing it into suitable formats, such as splitting it into training and validation sets.
3. FR3: The system should be able to train the selected models on the prepared dataset to analyze glacier dynamics and their impacts.
4. FR4: The system should be able to monitor the training progress by tracking parameters such as loss and accuracy to ensure effective learning.

2.2.3 Module 3

Following are the requirements for module 3:

1. FR1: System should compute the trained models on some performance metrics like accuracy, precision, etc.; also provide some visuals to compare models easily and store the results for future use.
2. FR2: System should be able to analyze errors as how much given predicted data is different from the actual label ones and then allow some parameter tuning or some adjustments in order to adjust errors.
3. FR3: System should be able to generate a report to compare each model in terms of accuracy, precision, computational efficiency.

2.2.4 Module 4

Following are the requirements for module 4:

1. FR1: The web app should be appealing, responsive to the user. The user can access targeted glaciers, for example Shisper Glacier, to see:
 - Satellite images after 2020
 - Classified images (colored labels)
 - Predictions (red/green zones)

- Reports for comparative analysis
2. FR2: The system should be able to map output georeferenced satellite images onto the actual Google Earth map. For testing purposes, the response time of the app should be fast and available on desktop and web.

2.3 Non-Functional Requirements

2.3.1 Reliability

- Requirement: The GlaciArc system shall maintain a Mean Time Between Failures (MTBF) of at least 1,000 hours.
- Failure Definition: A failure occurs when the system crashes, produces erroneous results, or experiences significant processing delays.
- Consequences: Software failures may lead to inaccurate predictions about glacier movements and GLOFs, adversely affecting disaster management decisions.
- Failure Protection: Regular data backups and performance logging will be implemented to safeguard against data loss.
- Error Detection Strategy: The system will employ robust exception handling to capture runtime errors and notify system administrators.
- Correction Strategy: A rollback mechanism will be in place to restore the system to its last stable state in the event of a failure.

2.3.2 Usability

- Requirement: Users shall be able to perform common tasks (e.g., downloading images, viewing predictions) within three clicks.
- Learning Ease: New users should comprehend basic functionalities within 30 minutes of interaction.
- Error Avoidance : The system shall feature an undo functionality for critical actions (e.g., deleting predictions).

2.3.3 Performance

- Requirement: The image preprocessing module must process images within 5 minutes.
- Responsiveness: The system shall respond to user interactions within 5 seconds for 95% of operations.
- Data Handling: The prediction module shall manage historical movement data from at least 5 years.

2.3.4 Security

- Requirement: The GlaciArc system shall ensure data protection by preventing unauthorized access.
- Access Control: The system shall support role-based access control, allowing different permission levels for users based on their roles.

2.4 Domain Model

This domain model diagram illustrates the data flow and processing steps involved in predicting glacial lake outburst floods (GLOFs). It shows how satellite images are processed, used to train a GLOF prediction model, and how the model's predictions are combined with infrastructure data to assess potential risks. The final output is visualized on an interactive map.

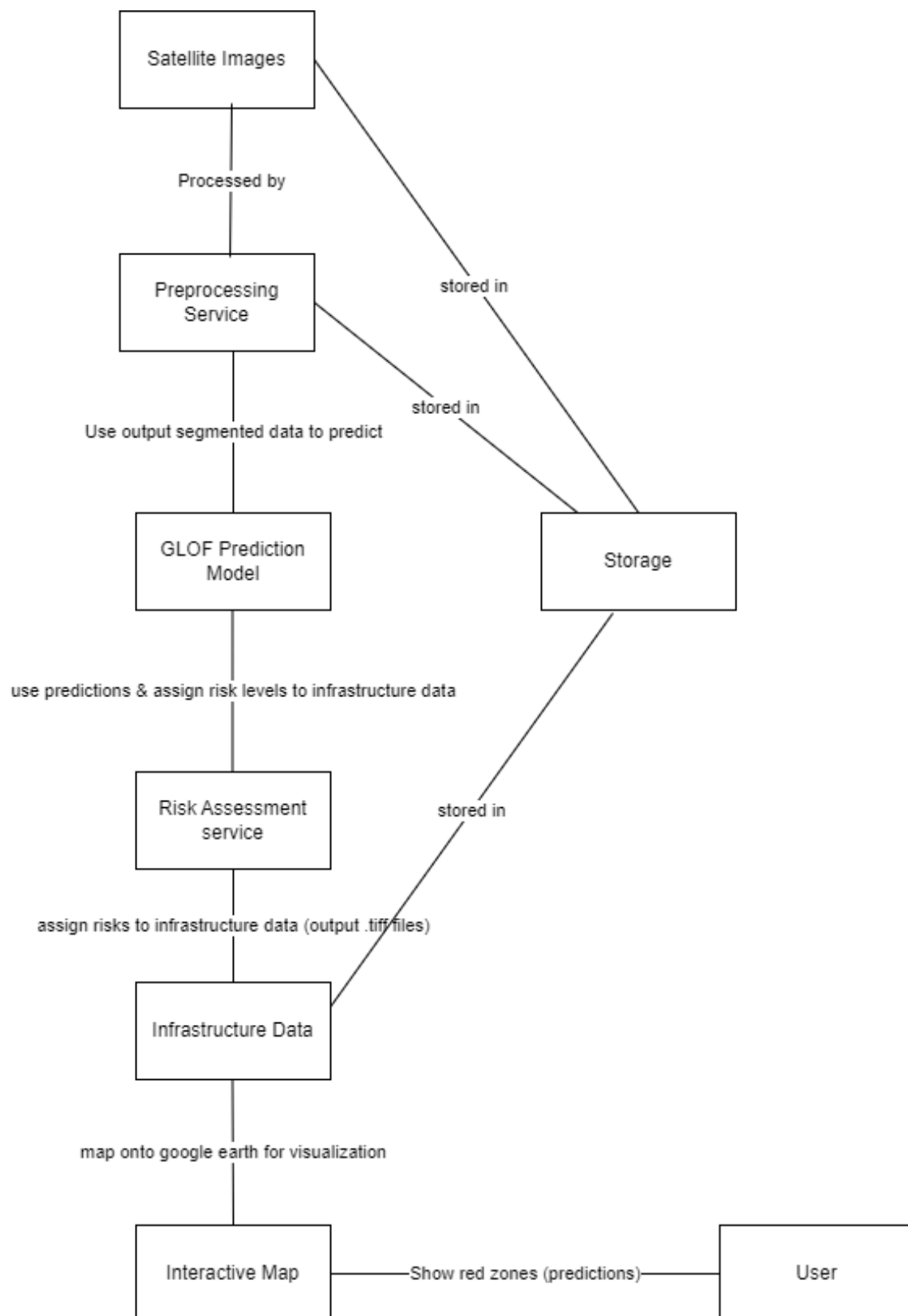


Figure 2.2: Domain model

2.5 Testing Plan for Iteration

2.5.1 1. Dataset Verification

- **Image Quality Assessment:** We assessed the resolution and clarity of the acquired satellite images to ensure that they are suitable for analysis.
- **Geographical Relevance:** We verified that the images obtained from Google Earth Engine cover the intended areas of Shisper and Passu glaciers and their surroundings.

2.5.2 2. Preprocessing Assessment

- **Verification of Preprocessing Steps:** We verified that all preprocessing steps, such as cropping, have been accurately executed.
- **Label Verification:** We ensured that the training dataset is correctly labeled, with appropriate class distributions for glaciers, water, houses, and other surrounding features.

2.5.3 3. Model Selection

We conducted research to determine the most suitable model for our project. After reviewing several papers, we concluded that the U-Net model is the optimal choice due to its effectiveness in segmenting images with complex structures, such as glaciers and their surroundings. The U-Net architecture is particularly known for its capability to deliver precise segmentation, making it ideal for our project.

Chapter 3

System Overview

3.1 Architectural Design

Our project follows a procedural programming structure, where each task is executed in a sequential manner. The workflow is based on a pipeline architecture, where the output of one step serves as the input for the next. The process begins with satellite image acquisition, downloading images from Google Earth Engine. These images are then pre-processed, followed by classification into categories such as glacier, infrastructure, water body, and vegetation. The segmented images are compared against ground truth values. After that the images are analyzed for glacier movement and other critical features to assess potential risks. Once the analysis is complete, the images are annotated with danger and safe zones. Finally, the results are visualized on the dashboard, providing a comprehensive view of the risks.



Figure 3.1: Box and line diagram of project architecture

3.2 Design Models

3.2.1 System Sequence Diagram

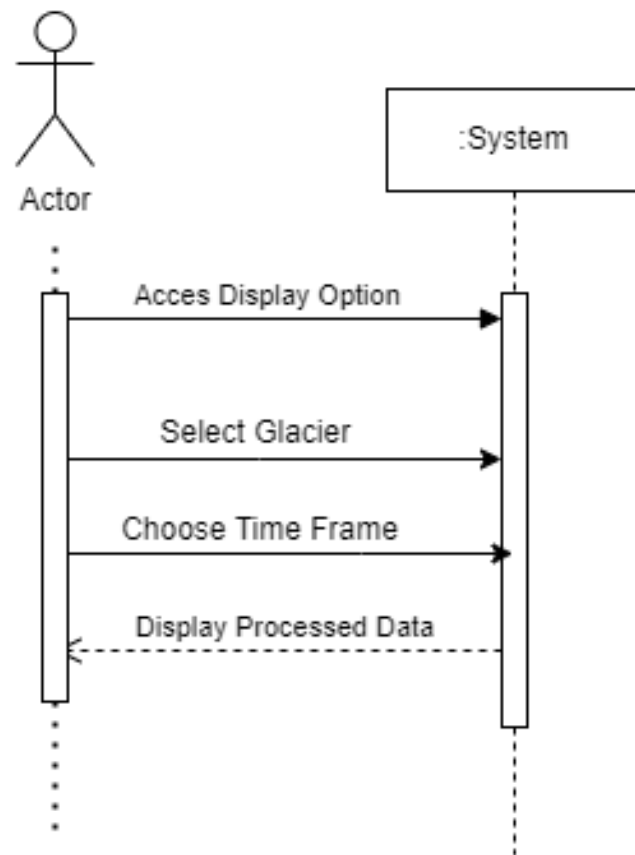


Figure 3.2: System Sequence Diagram for specific year selection

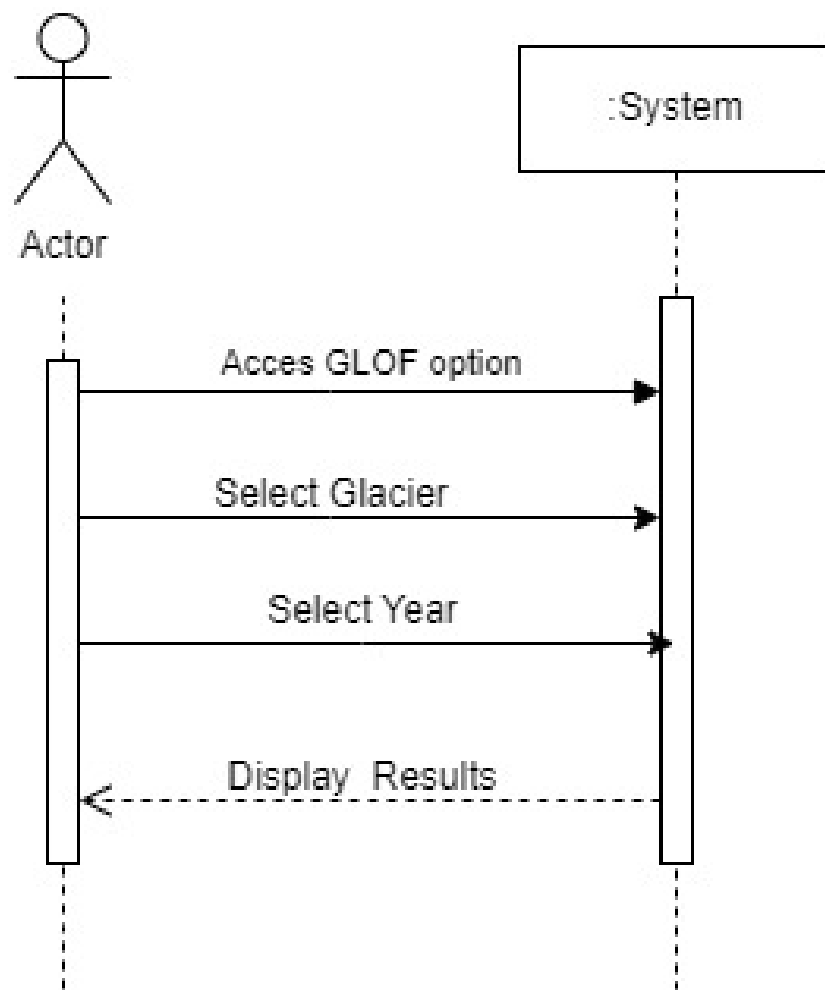


Figure 3.3: System Sequence Diagram for GLOF prediction

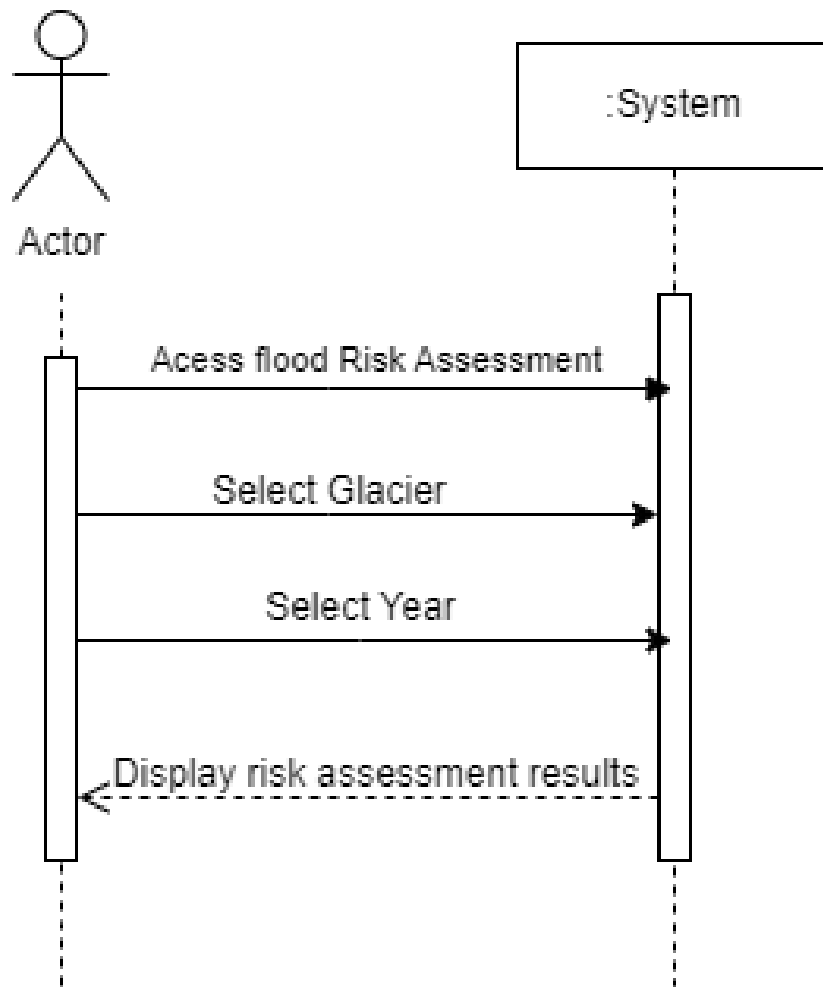


Figure 3.4: System Sequence Diagram for displaying risk zones

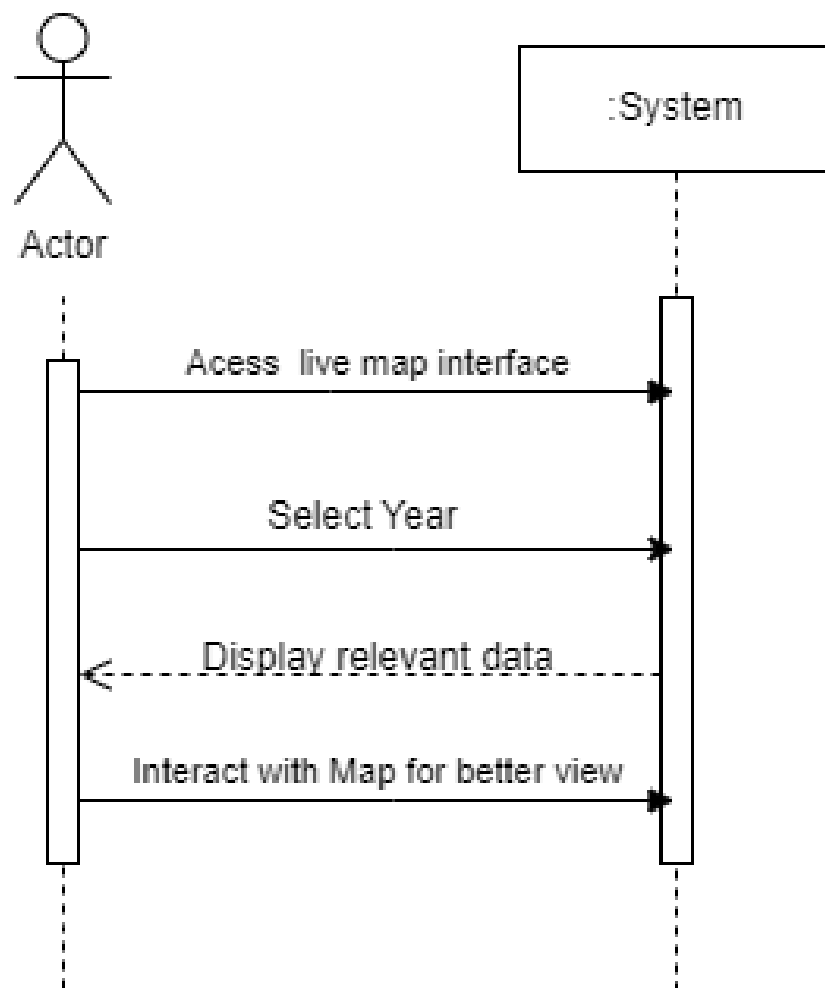


Figure 3.5: System Sequence Diagram for displaying interactive maps

3.2.2 Sequence Diagram

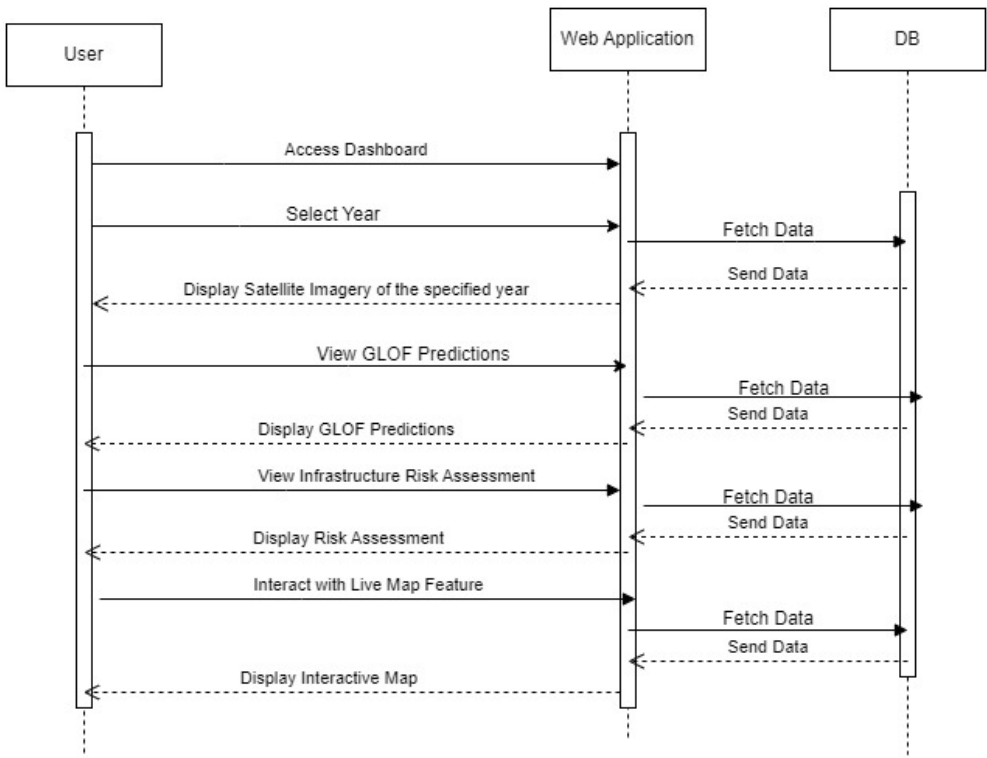


Figure 3.6: Sequence Diagram

3.2.3 Activity Diagram

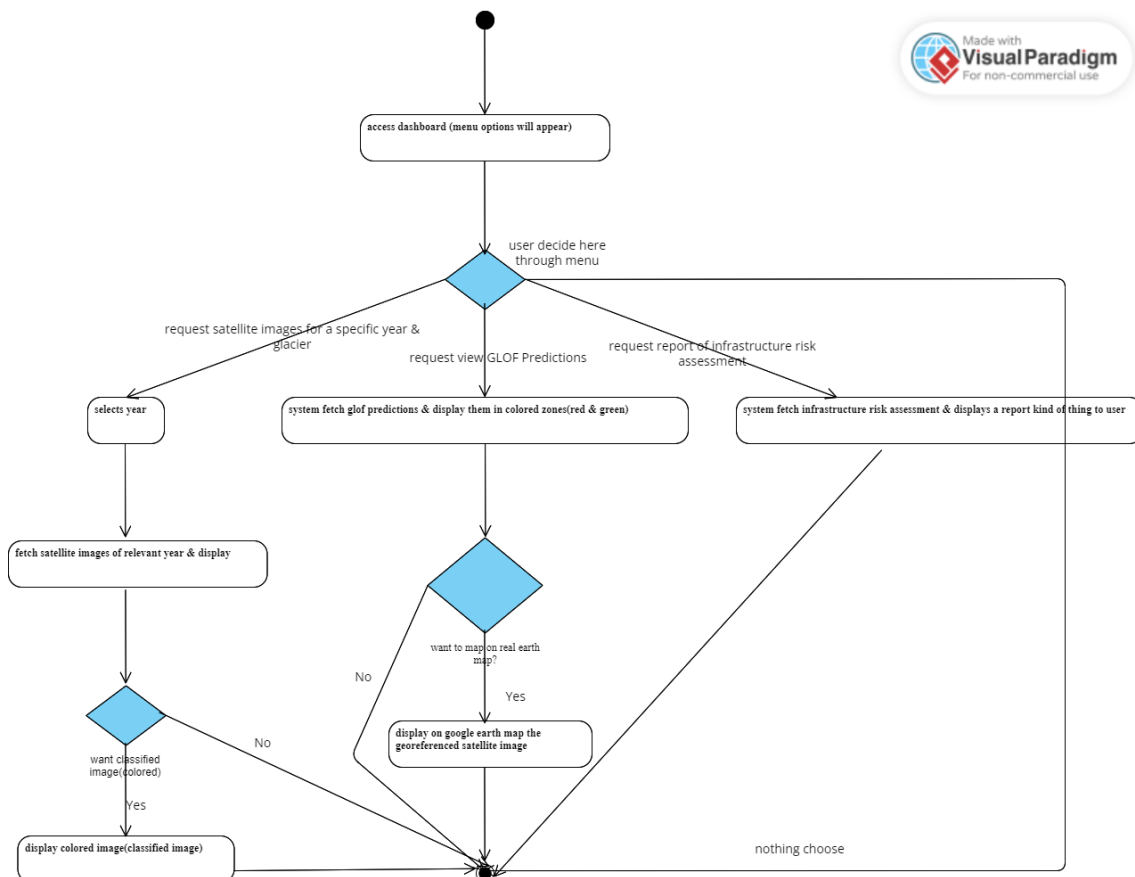


Figure 3.7: Activity Diagram

3.2.4 Data Flow Diagram

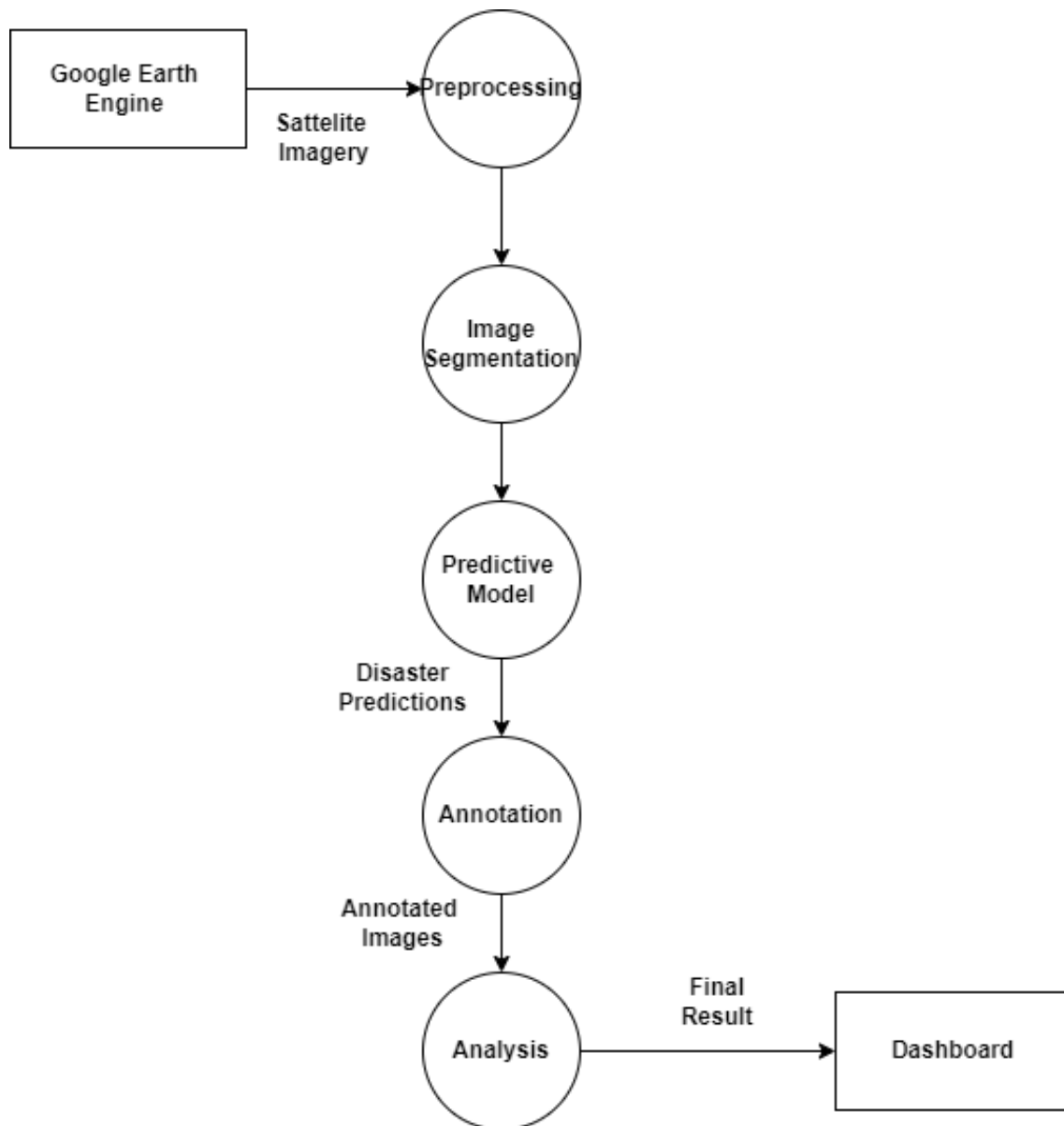


Figure 3.8: Data Flow Diagram

3.2.5 State Machine Diagram

This state machine diagram outlines the user flow for a system that assesses infrastructure risk from glacial lake outburst floods (GLOFs). Users can interact with a live map, view satellite imagery and risk assessments, access a dashboard to select years for analysis, and view predicted GLOF events. The system fetches relevant geospatial data and risk assessments based on user input.

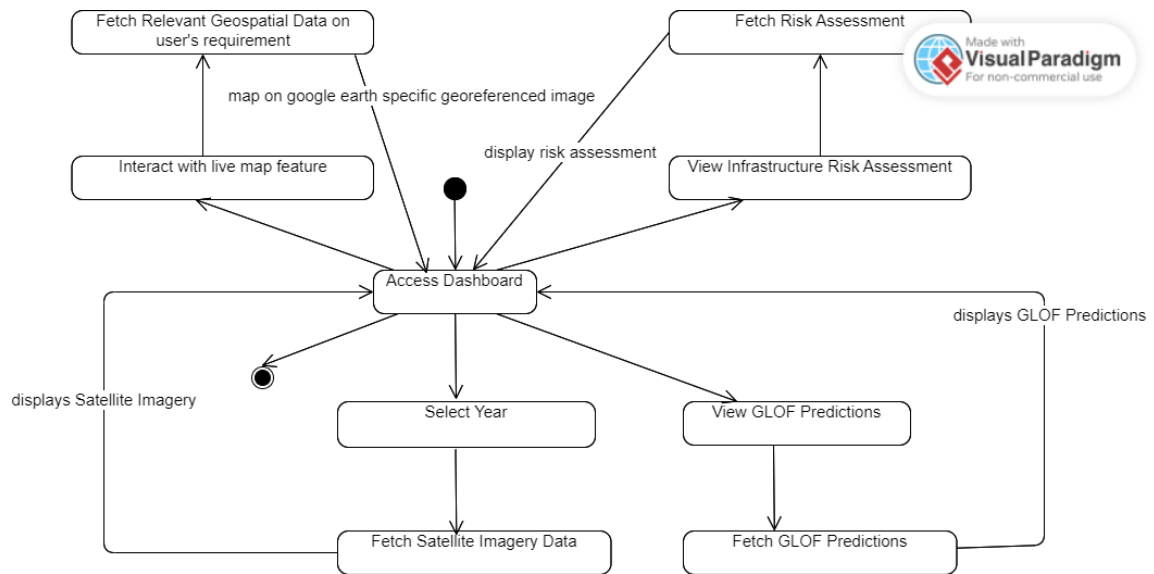


Figure 3.9: State Machine Diagram

3.2.6 Deployment Diagram

This deployment diagram depicts the architecture of a web application system. The user interacts with the system through a web browser, which communicates with a web server. The web server processes requests, interacts with an ML Model Server for data analysis, and stores results in a database. The system also leverages external services for additional data and functionality.

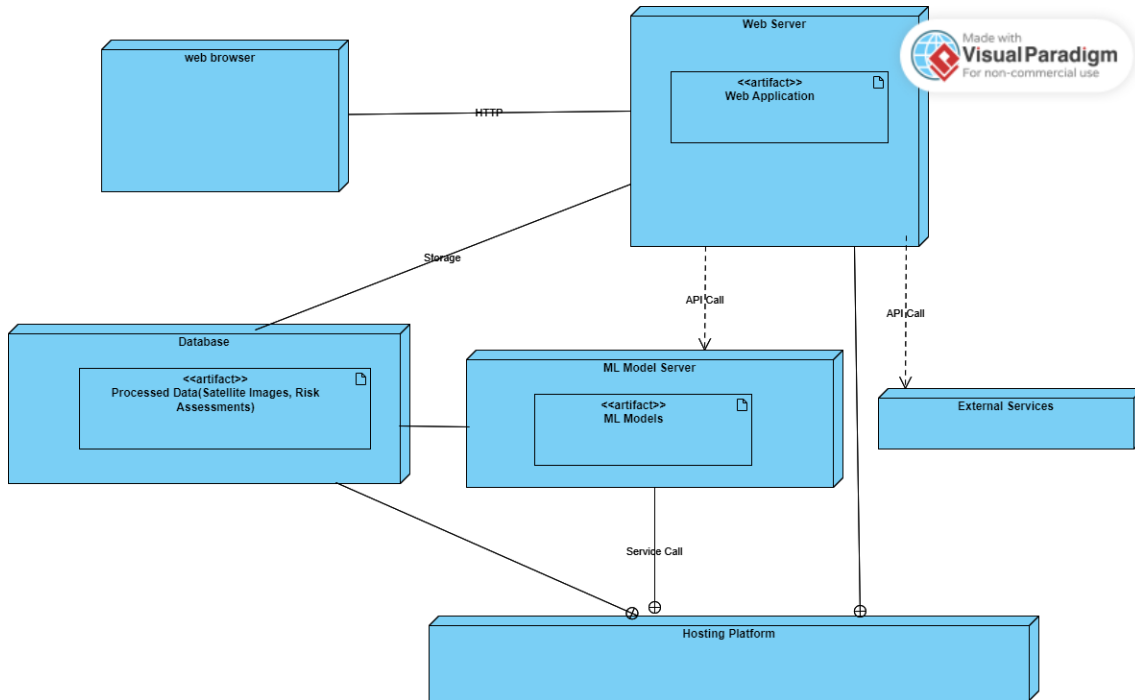


Figure 3.10: Deployment Diagram

3.3 Data Design

In the system, the satellite images are processed and stored on a cloud-based platform, ensuring scalability and secure access. Predictions regarding glacier movements and potential GLOFs are saved as annotated images, with safe and danger zones clearly marked on these images. These annotated images, along with their metadata, are accessible to users via the dashboard. User profiles are stored in a secure relational database.

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